Felipe **Cybis Pereira**

(+33) 6 02 17 62 62 · felipe.cybispereira@gmail.com · felipecybis.github.io

Research

Postdoctoral Researcher, neurosciences, functional ultrasound imaging — Paris, France
@ Physics for Medicine Paris, Inserm, ESPCI Paris, PSL, CNRS — 2025-2026
- Adviser: Sophie Pezet, PI, and Mickaël Tanter, PI.
- Focus: Data analysis for functional ultrasound imaging (fUSI) and Ultrasound Localization Microscopy (ULM).
- Techniques: **Data analysis for fUSI and ULM - Data Visualization - Python - General Linear Modeling - Skeletonization**

PhD, acoustics, neuroscience, spatial navigation and functional ultrasound imaging — Paris, France
@ Physics for Medicine Paris, Inserm, ESPCI Paris, PSL, CNRS — 2021-2025
- Adviser: Sophie Pezet, PI, and Mickaël Tanter, PI.
- Title: Revealing vascular patterns in the spatial navigation system with functional ultrasound imaging.
- Techniques: **Functional ultrasound imaging - Spatial navigation - Rodent's brain imaging - Animal behavior and cognition - Python - 3D design conception (CAD) - Ultrafast Ultrasound**
- Recognition: PariSanté Campus PhD Award 2025 (Innovation & Industry)

Research Intern, neurobiology and ethology — Boston, USA
@ Rogulja Lab, Harvard Medical School, Harvard University — 2019
- Adviser: Alexandra Vaccaro, PhD, and Dragana Rogulja, PI.
- Focus: Insights in sleep deprivation in *Drosophila melanogaster*.
- Techniques: ***Drosophila melanogaster*** **rearing - Immunostaining - Confocal microscopy - Ethological analysis - Survival and Dietary assays - Drosophila Activity Monitoring (DAM) system**

Teaching

Teacher assistant: Practical work in physiology and neuroscience — Paris, France
@ ESPCI Paris - PSL University — 2022-2024
- Professor: Thierry Gallopin, PhD.
- Neuroscience-focused practical work: **(1) human EEG (2) pose-estimation and animal tracking (3) human sleep**.

Teacher assistant: Methods for measuring and actuating neuronal activity — Saclay, France
@ M2 Computational Neurosciences and Neuroengineering, NeuroPSI — 2023-2025
- Professor: Isabelle Ferezou, PhD.
- Principles on Ultrafast Ultrasound, functional ultrasound imaging and Ultrasound Localization Microscopy.

Education

Doctorate Degree in Acoustics (applied to Neurosciences) — Paris, France
@ Université Paris Sciences et Lettres (PSL) — 2021-2025

Master's Degree in Bioengineering and Neurosciences — Paris, France
@ BioMedical Engineering Master — 2020-2021
- Scholarship student for the PSL Graduate Program in Life Sciences.

Engineering Degree (Biotechnology major) — Paris, France
@ ESPCI Paris - PSL — 2016-2019
- *Michelin Excellency* scholarship student in a double degree program with UFSC University in Brazil.

Bachelor's Degree in Chemical Engineering — Florianópolis, Brazil
@ Universidade Federal de Santa Catarina (UFSC) — 2014-2020

Skills

Programming
- **Scientific Python**: Neuroimaging (Nipy suite, BrainGlobe), Machine learning (scikit-learn), Visualization (VTK, PyVista).
- **General Python Packaging**: Publishing, documentation and examples, unit testing, linting and formatting, pre-commit hooks.
- **Version control**: Advanced usage of **Git**, **GitHub** and **GitHub Actions**.
- **Prototyping**: Intermediate knowledge in **Arduino** and **Raspberry Pi**.
- Intermediate knowledge on **Rust** (and **Rust bindings for Python**), **Lua**, **core-utils** and **shell-scripting**, **JavaScript**.
- Intermediate knowledge on text editing softwares such as LaTeX and **Typst**.

Spoken languages: **English** (fluent), **French** (fluent), **Portuguese** (native), Spanish (beginner)

Hobbies

I like playing **Tennis** (I played a lot while kid/teen), programming and tweaking my own dotfiles.

Publications and preprints

1. Khallaf, M.A., Hart, D.W., Luo, W., Murad, F., **Cybis Pereira, F.**, Mendez-Aranda, D., Hagenah, N., Rossi, A., Bégay, V., Okrouhlík, J., et al. (2026). A Queen Odour Mediates Reproductive Suppression in a Eusocial Mammal. *Nature*. https://doi.org/10.1038/s41586-026-10772-5.
2. Le Meur-Diebolt, S., **Cybis Pereira, F.**, Mariani, J.-C., Kliewer, A., Farinha-Ferreira, M., Bertolo, A., Osmanski, B.-F., Lenkei, Z., and Deffieux, T. (2026). Robust Functional Ultrasound Imaging in the Awake and Behaving Brain: A Systematic Framework for Motion Artifact Removal. *Imaging neuroscience (Cambridge, Mass.)*, *4*. https://doi.org/10.1162/IMAG.a.1191.
3. **Cybis Pereira, F.**, Castedo, S.H., Meur-Diebolt, S.L., Ialy-Radio, N., Bhattacharya, S., Ferrier, J., Osmanski, B.F., Cocco, S., Monasson, R., Pezet, S., et al. (2026). A Vascular Code for Speed in the Spatial Navigation System. *Cell Reports*, *45*. https://doi.org/10.1016/j.celrep.2025.116791.
4. Zucker, N., Le Meur-Diebolt, S., **Cybis Pereira, F.**, Baranger, J., Hurvitz, I., Demené, C., Osmanski, B.-F., Ialy-Radio, N., Biran, V., Baud, O., et al. (2025). Physio-fUS: A Tissue-Motion Based Method for Heart and Breathing Rate Assessment in Neurofunctional Ultrasound Imaging. *eBioMedicine*, *112*, 105581. https://doi.org/10.1016/j.ebiom.2025.105581.